

Diagonal-Translation Rigidity and Phase Reciprocity in Positive Newton-Fan Hamiltonian Shears

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Abstract

We study a proof-certified degree mechanism for compositions of separated Hamiltonian shears. Let the coefficient field have characteristic zero, let $r \geq 3$, and let the collected supports of the two potentials be finite, nonempty subsets of $\mathbb{Z}_{\geq 2}^r$. A strict edge records unique exposed support selectors, positive fresh carries, and the target certificates needed for the next phase. For every such edge, a grouped positive-face Hessian argument and the Jacobian criterion give coefficient-uniform survival of the exposed gradients under injective substitution. The leading phase is therefore the typed linear action $(u, w) \mapsto (B_\beta A_\alpha u, A_\alpha u)$, with $A_\alpha = \mathbf{1}\alpha^\top - I$ and $B_\beta = \mathbf{1}\beta^\top - I$. The two-phase degree vector translates in the all-ones direction. On every infinite certified strict branch, the total-degree envelopes have at most $d_V + d_W - 2$ selector changes, where d_V, d_W count distinct support totals, and the stationary tail is the global-origin affine recurrence $t_{n+1} = \lambda t_n + \mu$ with $\lambda = (|\alpha| - 1)(|\beta| - 1)$. Coordinate reversal yields a phase-resolved inverse reciprocity criterion on observable spans; full matrices occur only for full spans and only with reflected certificates. A four-projection margin lemma is proved for one typed pair core and one forward/reflected-inverse step. An exact asymmetric three-dimensional fixture and two failure fixtures delimit the assumptions. All statements are certified-branch results; no empirical, numerical, or computational evidence is used.

1 Introduction and contribution boundary

A pair of nonlinear shears can look deceptively simple when written in canonical coordinates. Let $q, p \in K^r$ and consider

$$S_V(q, p) = (q, p + \nabla V(q)), \quad T_W(q, p) = (q + \nabla W(p), p).$$

Each map is triangular, but the leading weighted degrees of an iterate are not determined by one scalar degree in general. A support exponent can win at one phase, a different exponent can win after the fresh block is substituted, and the inverse reads the phases in the opposite order. The resulting selector word is a finite-state object coupled to a growing degree vector. The purpose of this article is to isolate a regime in which that coupling has a rigid, auditable form.

The useful hypothesis is positivity of every support coordinate together with a strict certificate at the edge being followed. Positivity makes all support totals at least $2r$, but it does not by itself remove ties or cancellation. We therefore separate three logically different tasks. First, a grouped Hessian determinant supplies noncancellation for every nonzero coefficient choice, and a Jacobian argument transfers this independence through a fresh substitution. Second, strict carries make the formal degree action an actual leading-form action. Third, support functions along the resulting diagonal ray reduce the selector problem to upper envelopes of lines indexed by total degree. Keeping these tasks separate is what permits exact failure boundaries.

The main contributions are four bounded statements.

1. For every certified strict edge, the exposed gradient blocks survive uniformly over the coefficient class and the forward phase is $(u, w) \mapsto (B_\beta A_\alpha u, A_\alpha u)$. The inverse is obtained in W-first, V-second order, with subtraction signs.
2. Every certified two-phase edge translates u by a positive multiple of **1**. Along an infinite certified strict branch, equal-total walls are invariant and unequal-total walls are crossed at most once, giving the bound $d_V + d_W - 2$. Once the selector pair is stationary, the exact affine tail uses the original seed as its origin.
3. Coordinate reversal gives a necessary and sufficient phase-resolved reciprocity test on the spans actually observed by the certified integer cell. A proper span is not silently promoted to a full matrix identity.
4. A compact, normalized four-projection margin condition preserves one forward and one reflected inverse leading step on one typed pair core. It deliberately stops short of target-core inclusion and all-iterate perturbation stability.

The hypotheses are not presentation choices. If an exponent has a zero or unit coordinate, the determinant witness can vanish. If a carry is nonpos-

itive, an old block can cancel a fresh block. If a reflected support certificate is missing, the first inverse phase can already disagree. The exact examples in Sections 7 and 8 make these boundaries visible.

The article is organized as follows. Section 2 positions the mechanism against polynomial automorphisms, Newton and tropical degree methods, Hamiltonian shears, and dynamical-degree work. Section 3 fixes typed cells and states the omnibus theorem. Sections 4 and 5 prove survival, transport, translation, and the wall bound. Section 6 treats reversal and observable spans. Section 7 proves the local margin lemma and gives an exact fixture. Section 8 records limitations and a reproducibility statement.

Symbolic phase pipeline. A certified V selector α acts on u and produces the fresh vector $v = A_\alpha u$. A certified W selector β acts on v and produces $u' = B_\beta v$. The strict carries place each fresh block above the old block, so $u' = u + \delta \mathbf{1}$. Reversal starts from (Rw, Ru) , applies the reflected W label first, and then the reflected V label. This diagram is a notation aid, not an additional hypothesis.

2 Related work and claim-level collision positioning

The surrounding literature supplies several adjacent languages but not the conjunction used here. Reversors and polynomial symmetries in the plane provide terminology for phase reversal and fixed sets [12]; filtrations for polynomial automorphisms provide a broad forward/backward escape viewpoint [18]. Neither reference imposes positive collected Newton supports in dimension at least three, nor do they give the typed phase action proved below.

A second family studies monomial degree growth and its linear recurrences. The degree sequences and entropy questions for monomial maps [14] and matrix-induced recurrences [2] are useful contrasts. Polytope-algebra and mixed volume methods sharpen this toric perspective [10]. Our envelope variable is instead a support function along a diagonal ray, and the affine tail is conditional on a certified stationary selector pair. There is no claim that our matrices compute a global dynamical degree.

Symplectic and Hamiltonian contexts form a third family. Cluster-algebra constructions exhibit symplectic birational maps with tropical recurrences [11], while polynomial Hamiltonian flows with special straight-line orbits are studied in [16]. Structural results on polynomial symplectomorphisms [15] and shear generation for Hamiltonian maps [3] explain why

Table 1: Methodological adjacency and the missing conjunction.

Family	What is used as context	What is not inferred here
Polynomial reversors	phase order, symmetry vocabulary [12, 18]	no global reversor or plane reduction
Monomial/Newton	exposed faces, envelopes, recurrences [14, 2, 10, 9, 13]	no additive shear is replaced by a monomial map
Hamiltonian/symplectic	canonical shears and symplectic Jacobians [11, 16, 15, 3]	no generation, classification, or global Hamiltonian theorem
Dynamical degrees	degree-growth and spectral caution [5, 7, 4, 19]	no entropy or cohomological spectral identification
Recent tropical/dynamical	finite windows and current adjacency [20, 17, 1, 8, 6]	no priority claim and no transfer across map classes

separated shears are a natural laboratory. The present result is narrower: it is an exact leading-form statement for collected positive supports, not a classification of Hamiltonian maps or a generation theorem.

Dynamical-degree work for affine-triangular automorphisms [5], spectral interpretations [7], and monotonicity for Hénon-like families [4] illustrates why degree growth can have global geometric structure. Newton and tropical containment tools [9, 13] motivate our use of exposed faces and strict polyhedral certificates. Recent dimension-four affine-triangular results [19] and tropical degree work [17] are current adjacent contexts, but use different map classes and do not supply a reflected phase state.

Orbit graphs and finite-window functoriality [20], as well as relative dynamical degrees for twisted rational maps [1], are useful negative controls: a finite selector word here is certified edge by edge and does not automatically close under an ambient graph or a rational fibration. Higher-dimensional degree-growth examples [8] and general inverse-degree bounds [6] likewise do not imply the exact inverse reciprocity in Section 6.

The public literature check underlying these comparisons is bounded through 2026-08-29 UTC. It is a bounded public screen rather than an exhaustive survey or a priority statement. The closest mechanisms are summarized in Table 1.

3 Typed setup, strict cells, and omnibus theorem

Let K be a field of characteristic zero and let $r \geq 3$. Let

$$E_V, E_W \subset \mathbb{Z}_{\geq 2}^r$$

be finite, nonempty, collected supports. Write

$$V(q) = \sum_{\alpha \in E_V} c_\alpha q^\alpha, \quad W(p) = \sum_{\beta \in E_W} d_\beta p^\beta, \quad c_\alpha, d_\beta \in K^\times.$$

The variables and incoming leading tuples used for degree bookkeeping are in disjoint blocks. A degree vector $u \in \mathbb{R}_{>0}^r$ assigns $\deg_u(q^\alpha) = \alpha \cdot u$. A seed consists of positive integer vectors and algebraically independent incoming leading blocks; disjoint fresh blocks realize every declared integer seed.

Define the shears and their exposed actions by

$$S_V(q, p) = (q, p + \nabla V(q)), \quad T_W(q, p) = (q + \nabla W(p), p), \quad F = T_W \circ S_V,$$

$$A_\alpha = \mathbf{1}\alpha^\top - I, \quad B_\beta = \mathbf{1}\beta^\top - I.$$

The support functions are

$$h_V(u) = \max_{\alpha \in E_V} \alpha \cdot u, \quad h_W(v) = \max_{\beta \in E_W} \beta \cdot v.$$

For a label pair $e = (\alpha, \beta)$, define the u-only cone

$$\mathcal{K}_e^u = \{u \in \mathbb{R}_{>0}^r : \alpha \text{ is the unique } V \text{ maximizer at } u, \beta \text{ is the unique } W \text{ maximizer at } A_\alpha u, B_\beta A_\alpha u - u\}$$

The source fresh carry and the pair-dependent predicates are kept separate:

$$\mathcal{C}_e^+ = \{(u, w) \in \mathbb{R}_{>0}^r \times \mathbb{R}_{>0}^r : u \in \mathcal{K}_e^u, \quad A_\alpha u - w > 0, \quad \text{PairGaps}_e(u, w) > 0\}. \quad (3.1)$$

All vector inequalities are componentwise. The integer cell and observable spans are

$$\mathcal{C}_e^{\mathbb{Z}} = \mathcal{C}_e^+ \cap (\mathbb{Z}_{>0}^r)^2, \quad U_e = \text{span}_{\mathbb{R}}\{u : (u, w) \in \mathcal{C}_e^{\mathbb{Z}}\}, \quad V_e = \text{span}_{\mathbb{R}}\{A_\alpha u : (u, w) \in \mathcal{C}_e^{\mathbb{Z}}\}. \quad (3.2)$$

The reflected cell is $R_{\text{state}}(\mathcal{C}_e^+)$ and its pair predicates are checked after the swap; they are not folded into $\mathcal{K}_e^u$.

Definition 3.1 (Certified strict branch). A finite or infinite edge word is certified when every edge has unique selectors, positive source and target carries, a nonempty integer cell, and the printed reflected score, carry, and target certificates required for the phases being asserted. A finite word may stop at a tie, failed carry, empty cell, or missing certificate. The phrase "infinite certified strict branch" always includes all these certificates at every edge.

Theorem 3.2 (Certified phase, translation, and reciprocity theorem). *For the supports and shears above, the following assertions hold.*

- (A) *For every certified strict edge and every nonzero coefficient choice in the stated class, the grouped positive-face Hessian is nonzero and its face-gradient components are algebraically independent. Injective substitution and strict carries give the forward leading action*

$$(u, w) \mapsto (B_\beta A_\alpha u, A_\alpha u).$$

The inverse uses W first and V second, with subtraction signs.

- (B) *On every certified edge,*

$$v = h_V(u)\mathbf{1} - u, \quad u' = h_W(v)\mathbf{1} - v = u + \delta(u)\mathbf{1}, \quad \delta(u) > 0.$$

On every infinite certified strict branch, the number of selector changes is at most $d_V + d_W - 2$, where d_V, d_W count distinct support totals. After a stationary selector pair is reached,

$$t_{n+1} = \lambda t_n + \mu, \quad \lambda = (|\alpha| - 1)(|\beta| - 1), \quad \mu = ((|\beta| - 1)\alpha - \beta) \cdot u_0,$$

with $u_n = u_0 + t_n\mathbf{1}$ and the global origin $t_0 = 0$.

- (C) *With coordinate reversal R and $R_{\text{state}}(u, w) = (Rw, Ru)$, reflected inverse reciprocity is necessary and sufficient edgewise on U_e, V_e , provided reflected selector, carry, and target certificates are present. Full-matrix identities are asserted only when the corresponding spans are full.*

- (D) *If a single typed pair core has the four normalized projections and selected-minus-new margins of Lemma 7.1, then exactly one forward and one reflected inverse leading step are preserved. No target-core inclusion or all-iterate stability is asserted.*

4 Positive-face survival and degree transport proofs

We begin with the canonical geometry, because the phase order matters. The Jacobians are

$$DS_V = \begin{pmatrix} I & 0 \\ \text{Hess } V & I \end{pmatrix}, \quad DT_W = \begin{pmatrix} I & \text{Hess } W \\ 0 & I \end{pmatrix}.$$

The Hessians are symmetric. With $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$, direct multiplication gives

$$DS_V^T J DS_V = J, \quad DT_W^T J DT_W = J.$$

Thus S_V, T_W , and F are polynomial symplectomorphisms. Their inverses are

$$S_V^{-1}(q, p) = (q, p - \nabla V(q)), \quad T_W^{-1}(q, p) = (q - \nabla W(p), p),$$

so

$$F^{-1} = S_V^{-1} \circ T_W^{-1}. \quad (4.1)$$

The inverse order is W first and V second; the minus signs are nonzero scalars in K .

Let E_0 be an exposed face of either support and set

$$P_{E_0}(X) = \sum_{\alpha \in E_0} c_\alpha X^\alpha, \quad h_{ij}(\alpha) = \alpha_i(\alpha_j - \delta_{ij}).$$

Grouping the determinant by row-labelled tuples gives

$$\det \text{Hess } P_{E_0} = \sum_{(\alpha_1, \dots, \alpha_r) \in E_0^r} \left(\prod_i c_{\alpha_i} \right) D(\alpha_1, \dots, \alpha_r) X^{\sum_i \alpha_i - 2\mathbf{1}}, \quad (4.2)$$

where $D(\alpha_1, \dots, \alpha_r) = \det[h_{ij}(\alpha_i)]$. Choose a generic secondary linear weight on E_0 with a unique minimizer α_0 . The tuple $(\alpha_0, \dots, \alpha_0)$ is the unique lowest secondary-weight group. Its matrix is

$$[h_{ij}(\alpha_0)] = \text{diag}(\alpha_0)(\mathbf{1}\alpha_0^T - I).$$

The matrix determinant lemma therefore yields

$$D(\alpha_0, \dots, \alpha_0) = (-1)^r (1 - |\alpha_0|) \prod_i \alpha_{0i} \neq 0. \quad (4.3)$$

The associated coefficient is $c_{\alpha_0}^r (-1)^r (1 - |\alpha_0|) \prod_i \alpha_{0i}$. It cannot vanish: the field has characteristic zero, every coefficient is nonzero, and every support coordinate is at least two. Thus the grouped determinant is nonzero for every coefficient choice in the class. The argument covers a one-point face and a tied face. In the tied case it is only an auxiliary wall certificate; it does not select a tied vertex.

The characteristic-zero Jacobian criterion now implies that the components of ∇P_{E_0} are algebraically independent. If $Z_1, \dots, Z_r$ are independent incoming leading forms, the substitution $X_i \mapsto Z_i$ is injective on the polynomial algebra generated by the face tuple. A selected face gradient therefore cannot disappear through a hidden coefficient cancellation. For a strict singleton this is also visible from

$$\det A_\alpha = (-1)^r (1 - |\alpha|) \neq 0. \quad (4.4)$$

Suppose α is the unique V selector at u and β is the unique W selector at $A_\alpha u$. Each component of ∇V has fresh weighted degree $A_\alpha u$. The first strict carry

$$A_\alpha u - w > 0 \quad (4.5)$$

places this block above the old p block. The W gradient then has degree $B_\beta A_\alpha u$, and the second carry

$$B_\beta A_\alpha u - u > 0 \quad (4.6)$$

places it above the old q block. Algebraic independence prevents cancellation inside a fresh block, while (4.5)–(4.6) prevent cancellation with an old block. Consequently

$$(u, w) \mapsto (u', v) = (B_\beta A_\alpha u, A_\alpha u). \quad (4.7)$$

For the inverse, start at $R_{\text{state}}(u, w)$, use the reflected W label first, then the reflected V label, and multiply the fresh blocks by -1 . No nonzero scalar sign changes a strict leading degree. If a selector ties, a carry fails, or the integer cell is empty, the edge is not certified and no implicit equality is claimed.

A tied-face Hessian certificate deserves one further warning. It proves independence of the whole face gradient after a generic secondary specialization, but it does not choose one vertex on the primary wall. Selector uniqueness remains a separate strict hypothesis.

5 Translation, wall monotonicity, and affine tail proofs

Let $s = \alpha \cdot u = h_V(u)$ on a certified edge. The first phase gives

$$v = A_\alpha u = s\mathbf{1} - u = h_V(u)\mathbf{1} - u. \quad (5.1)$$

For the selected β ,

$$\beta \cdot A_\alpha u = |\beta|s - \beta \cdot u.$$

Hence

$$\begin{aligned} u' &= B_\beta v = (\beta \cdot v)\mathbf{1} - v \\ &= u + ((|\beta| - 1)\alpha - \beta) \cdot u \mathbf{1}. \end{aligned} \quad (5.2)$$

The support-function form of the increment is

$$\delta(u) = h_W(v) - h_V(u) = ((|\beta| - 1)\alpha - \beta) \cdot u > 0. \quad (5.3)$$

The final inequality is precisely the second strict carry. For an integer seed, all matrices are integral, so $\delta(u) \in \mathbb{Z}_{\geq 1}$. Iterating (5.1)–(5.3) gives

$$u_n = u_0 + t_n \mathbf{1}, \quad t_0 = 0, \quad t_{n+1} - t_n = \delta(u_n) > 0. \quad (5.4)$$

Thus the diagonal direction is an actual weighted-degree identity, not only a formal matrix property.

The selector count is controlled by total-degree classes. Put

$$T_V = \{|\alpha| : \alpha \in E_V\}, \quad T_W = \{|\beta| : \beta \in E_W\}, \quad d_V = |T_V|, \quad d_W = |T_W|.$$

Along $u(t) = u_0 + t\mathbf{1}$ define

$$M_s = \max_{\substack{\alpha \in E_V \\ |\alpha|=s}} \alpha \cdot u_0, \quad h_V(u(t)) = \max_{s \in T_V} (M_s + st),$$

and

$$g(t) = h_V(u(t)) - t = \max_{s \in T_V} (M_s + (s - 1)t). \quad (5.5)$$

Every slope $s - 1$ is positive. If $t_2 > t_1$, an argmax line at t_1 has strictly increased by $(s - 1)(t_2 - t_1)$, so $g(t_2) > g(t_1)$.

Equation (5.1) becomes

$$v(t) = g(t)\mathbf{1} - u_0. \quad (5.6)$$

For $\beta, \eta \in E_W$,

$$\beta \cdot v(t) - \eta \cdot v(t) = (|\beta| - |\eta|)g(t) - (\beta - \eta) \cdot u_0. \quad (5.7)$$

Thus W selectors form an upper envelope of d_W lines in the increasing variable g . A wall between unequal totals is crossed at most once. A wall between equal totals is invariant along the ray, because the first term in (5.7) vanishes. The V envelope has at most $d_V - 1$ changes and the W envelope at most $d_W - 1$, so an infinite strict phase word satisfies

$$\#\{\text{selector changes}\} \leq (d_V - 1) + (d_W - 1) = d_V + d_W - 2. \quad (5.8)$$

A discrete branch can skip a wall and only lower this count. A finite branch may terminate at a tie, failed carry, empty cell, or missing certificate; (5.8) is not applied beyond that stopping event.

After finitely many changes on an infinite branch, fix a stationary pair (α, β) and write

$$c = (|\beta| - 1)\alpha - \beta.$$

Since $c \cdot \mathbf{1} = (|\alpha| - 1)(|\beta| - 1) - 1$, (5.3) and (5.4) give

$$t_{n+1} = \lambda t_n + \mu, \quad \lambda = 1 + c \cdot \mathbf{1} = (|\alpha| - 1)(|\beta| - 1), \quad \mu = c \cdot u_0. \quad (5.9)$$

The origin is global: $t_0 = 0$ and u_0 is the original seed. Resetting the origin after a transient selector change produces a different intercept and is not the claimed formula. Positivity gives $|\alpha|, |\beta| \geq 2r$, hence $\lambda > 1$. No minimal scalar recurrence, Perron degree, entropy, or higher dynamical-degree conclusion follows from (5.9).

Envelope picture. The ray $u_0 + t\mathbf{1}$ is parameterized by t . The V envelope is the maximum of lines $M_s + st$; subtracting t gives $g(t)$ with positive slopes $s - 1$. The W envelope is the maximum of lines in the increasing variable $g(t)$. Equal-total walls are parallel to the ray and unequal-total walls are single-crossing.

6 Reflected inverse reciprocity and its span criterion

Let R reverse coordinates, so $R\mathbf{1} = \mathbf{1}$, $R^2 = I$, and $R^\top = R$. Write $\bar{\alpha} = R\alpha$ and $\bar{\beta} = R\beta$. A direct calculation gives

$$B_{\bar{\alpha}}R = (\mathbf{1}(R\alpha)^\top - I)R = \mathbf{1}\alpha^\top - R = RA_\alpha,$$

and similarly

$$A_{\bar{\beta}}R = RB_{\beta}. \quad (6.1)$$

For $v = A_{\alpha}u$ and $u' = B_{\beta}v$, starting at $R_{\text{state}}(u, w) = (Rw, Ru)$, the reflected inverse phase is

$$B_{\bar{\alpha}}Ru = RA_{\alpha}u = Rv, \quad A_{\bar{\beta}}Rv = RB_{\beta}v = Ru'. \quad (6.2)$$

The first step uses the reflected W label and the second the reflected V label. It is not the ordinary forward order with a permutation appended.

The observable spans in (3.2) encode exactly what integer seeds can test. We state the equivalence as a proposition.

Proposition 6.1 (Edgewise reciprocity). *Fix a certified edge $e = (\alpha, \beta)$. Assume the reflected score, carry, and target certificates hold on the reflected cell. The following are equivalent:*

- (i) *For every integer seed in $\mathcal{C}_e^{\mathbb{Z}}$, the phase-resolved one-step forward vector and the reflected inverse vector agree, and the equality persists along every edge of a finite certified seed-indexed word.*
- (ii) *The restrictions*

$$B_{\bar{\alpha}}R|_{U_e} = RA_{\alpha}|_{U_e}, \quad A_{\bar{\beta}}R|_{V_e} = RB_{\beta}|_{V_e} \quad (6.3)$$

hold on every edge of that word.

If $U_e = V_e = \mathbb{R}^r$, (6.3) upgrades to full-matrix identities. On a proper span it does not. Equality of scalar total degrees, or comparison of a different same numerical seed that is not fixed by R_{state} , is insufficient.

Proof. For necessity, restrict the first phase equality to all integer seeds in the cell. Their real span is U_e , so linearity forces the first identity in (6.3). The second phase equality applied to $A_{\alpha}u$ spans V_e and forces the second identity. For sufficiency, apply the first restriction to each edge seed and use the reflected carry and target certificates to identify the next typed cell. Apply the second restriction there. Induction over the finite edge-indexed word gives equality at every phase. The full-span statement is ordinary equality of linear maps; no such upgrade is possible from a proper span. $\square$

The proposition is deliberately edgewise. It does not assert a global support reflection, a global reversor, a conjugacy, or an equality of maps outside the certified cells. The missing-reflection example in Section 8 shows why the reflected certificate is part of the hypothesis.

7 Local lower-ideal lemma and exact fixture

We first formulate the one-step margin statement without pretending that a cone is a pair cell. Fix one strict typed pair core and choose a nonempty compact set

$$\widehat{\Sigma}_e^+ \subset \mathcal{C}_e^+ \cap \{\|(u, w)\|_1 = 1\}$$

on which every coordinate of $u, w, A_\alpha u - w$ is at least $\varepsilon > 0$, and every selected score, carry, target, and reflected pair gap is at least $\eta > 0$. Define the four normalized projections

$$\begin{aligned} X^{V,+} &= \pi_u(\mathcal{C}_e^+), & X^{W,+} &= \{A_\alpha u : (u, w) \in \mathcal{C}_e^+\}, \\ X^{W,-} &= \{Ru : (u, w) \in \mathcal{C}_e^+\}, & X^{V,-} &= \{RA_\alpha u : (u, w) \in \mathcal{C}_e^+\}. \end{aligned}$$

Let $\widehat{X}^{S,\pm}$ denote their closures after ℓ^1 normalization and the coordinate lower bound ε . For each retained row ℓ and lower/new row ρ in the associated section assume

$$\min_{x \in \widehat{X}^{S,\pm}} (\ell - \rho) \cdot x \geq \Delta_{S,\pm} > 0. \quad (7.1)$$

For an unnormalized $x = s\widehat{x}$, homogeneity makes the gap at least $s\Delta_{S,\pm}$. Typed pair margins supply the analogous inequalities involving w and Rw .

Lemma 7.1 (One-edge four-projection margin lemma). *Under the preceding compactness and margin assumptions, the selected rows $(\alpha, \beta, R\alpha, R\beta)$ remain selected, with positive carries, for exactly one forward step and exactly one reflected inverse step. The conclusion is a leading-form statement on the one typed pair core.*

Proof. Each selected-minus-new row is positive on the normalized closure by (7.1). Homogeneity transfers positivity to every scale in the core. The source and target pair margins keep the new vector in the certified inequalities for the next phase, while the reflected pair margins do the same after the state swap. The survival argument of Section 4 then turns these strict inequalities into leading-form selection for one step in each direction. No assertion is made about a second edge, because no target-core inclusion was assumed. In particular, a forward margin does not imply the reflected margin or any all-iterate perturbation stability. $\square$

We now give an exact asymmetric fixture. It supplies arithmetic witness, not universal evidence. Let

$$E_V = \{a_1 = (8, 2, 2), a_2 = (2, 5, 6), \gamma = (2, 8, 2)\}, \quad E_W = \{b_1 = (2, 2, 8), b_2 = (6, 5, 2)\},$$

with $b_i = Ra_i$ and $R\gamma \notin E_W$. The exposed matrices are

$$A_1 = \begin{pmatrix} 7 & 2 & 2 \\ 8 & 1 & 2 \\ 8 & 2 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 5 & 6 \\ 2 & 4 & 6 \\ 2 & 5 & 5 \end{pmatrix}, \quad A_\gamma = \begin{pmatrix} 1 & 8 & 2 \\ 2 & 7 & 2 \\ 2 & 8 & 1 \end{pmatrix},$$

$$B_1 = \begin{pmatrix} 1 & 2 & 8 \\ 2 & 1 & 8 \\ 2 & 2 & 7 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 5 & 5 & 2 \\ 6 & 4 & 2 \\ 6 & 5 & 1 \end{pmatrix}.$$

The u-only strict cones are

$$\begin{aligned} K_1 : & (6, -3, -4) \cdot u > 0, \quad (6, -6, 0) \cdot u > 0, \\ & (4, -1, 8) \cdot u > 0, \quad (4, 5, 2) \cdot u > 0; \\ K_2 : & (-6, 3, 4) \cdot u > 0, \quad (0, -3, 4) \cdot u > 0, \\ & (-2, 2, 12) \cdot u > 0, \quad (-2, 8, 6) \cdot u > 0. \end{aligned}$$

Set $\mathcal{C}_i^+ = \{(u, w) : u \in K_i, w > 0, A_i u - w > 0\}$ and $\mathcal{C}_i^- = R_{\text{state}}(\mathcal{C}_i^+)$. These are the corresponding positive and reflected pair cells. Direct multiplication gives

$$C_{21} = B_2 A_1 = I + \mathbf{1}(90, 19, 22), \quad C_{22} = B_2 A_2 = I + \mathbf{1}(18, 55, 70). \quad (7.2)$$

For the C1-to-C2 transition, the target and carry forms are

$$\begin{aligned} (a_2 - a_1)C_{21}u &= (84, 22, 26) \cdot u, \\ (a_2 - \gamma)C_{21}u &= (90, 16, 26) \cdot u, \\ (b_2 - b_1)A_2 C_{21}u &= (1078, 230, 276) \cdot u, \\ (a_2 - \gamma)RA_2 C_{21}u &= (1078, 236, 270) \cdot u, \\ A_2 C_{21}u - A_1 u &= \mathbf{1}(1074, 231, 268) \cdot u. \end{aligned} \quad (7.3)$$

For the C2 self-loop they are

$$\begin{aligned} (a_2 - a_1)C_{22}u &= (12, 58, 74) \cdot u, \\ (a_2 - \gamma)C_{22}u &= (18, 52, 74) \cdot u, \\ (b_2 - b_1)A_2 C_{22}u &= (214, 662, 852) \cdot u, \\ (a_2 - \gamma)RA_2 C_{22}u &= (214, 668, 846) \cdot u, \\ A_2 C_{22}u - A_2 u &= \mathbf{1}(216, 660, 840) \cdot u, \\ C_{22}u - u &= \mathbf{1}(18, 55, 70) \cdot u. \end{aligned} \quad (7.4)$$

The source W gap and reflected source gap are $(4, -1, 8) \cdot u$ and $(4, 5, 2) \cdot u$ in C1, and $(-2, 2, 12) \cdot u$ and $(-2, 8, 6) \cdot u$ in C2. All displayed forms are positive on their respective cones.

Table 2: Fixture data and scope.

Item	Exact value or interpretation
Supports	$E_V = \{(8, 2, 2), (2, 5, 6), (2, 8, 2)\}; \quad E_W = \{(2, 2, 8), (6, 5, 2)\}$
Products	$C_{21} = I + \mathbf{1}(90, 19, 22); \quad C_{22} = I + \mathbf{1}(18, 55, 70)$
Determinants	$\det A_1 = 11, \det A_2 = 12$; seed determinants are nonzero
C1 transition	Five positive forms in (7.3), including the target carry
C2 loop	Six positive forms in (7.4), including the diagonal increment
Reflected path	Identities (7.5), with local rather than global reflection
Scope	Arithmetic witness only; no universal proof or perturbation claim

The C1 seed $((2, 1, 1), (1, 1, 1))$ has V scores 20, 15, 14, gives $A_1 u = (18, 19, 19)$ and $C_{21} u = (223, 222, 222)$, and has W gap 15. The ordinary C2 seed $((1, 1, 1), (1, 1, 1))$ gives $A_2 u = (12, 12, 12)$ and $C_{22} u = (144, 144, 144)$; it is a stationary baseline, not a transient witness. Three independent C1 u-seeds are $(2, 1, 1), (2, 1, 2), (3, 1, 1)$, and three independent C2 u-seeds are $(1, 1, 1), (1, 1, 2), (1, 2, 2)$. Their determinants are nonzero, while $\det A_1 = 11$ and $\det A_2 = 12$, so the corresponding V spans are full.

The reflected arithmetic is

$$B_1 R u = R A_1 u, \quad A_2 B_1 R u = R C_{21} u, \quad B_2 R u = R A_2 u, \quad A_2 B_2 R u = R C_{22} u. \quad (7.5)$$

It is local: at $u = (1, 10, 1)$, $\gamma \cdot u = 84$, while $a_1 \cdot u = 30$ and $a_2 \cdot u = 58$, and $R\gamma$ is absent from E_W . Thus the fixture is not a map-level reversor.

8 Boundaries, limitations, reproducibility, and conclusion

The hypotheses can fail in several independent ways. If an exponent has a zero or unit coordinate, the coefficient in (4.3) can vanish or the fresh all-ones carry can lose positivity. If a strict selector becomes a tie, the face certificate does not choose a vertex. If $A_\alpha u \leq w$ or $\delta(u) \leq 0$, an old block can remain visible and (5.4) is not an actual degree law. Empty integer cells and missing reflected target certificates are stopping boundaries, not cases where an equality is silently extended.

A cancellation fixture makes the role of the class concrete. Take

$$V = (q_1 + q_2 + q_3)^3, \quad W = (p_1 - p_2)^3. \quad (8.1)$$

The support contains zero and unit coordinates, and the leading V-gradient components are equal. At the W half-step of the first iterate, the leading $p_1 - p_2$ terms cancel, so a degree predicted by an unqualified fresh-block rule is lost. This example lies outside the headline class and is not a claim about a second iterate.

The reflection hypothesis has its own failure. If $R\gamma$ is absent from the appropriate reflected support or action certificate, the identities (6.2) cannot be invoked. On an open integer seed cell this produces an $n = 1$ phase-vector mismatch. Likewise, a proper observable span supports only the restricted equalities (6.3), not a full matrix assertion. Scalar total-degree equality and a comparison of unrelated same numerical seeds cannot repair either defect.

For clarity, the article makes no claim of arbitrary-support or zero-coordinate extension, an unconditional selector algorithm, a global map-level reversor, global conjugacy, classification, entropy, higher dynamical degrees, a minimal scalar recurrence, a Perron algebraic degree, same-seed reciprocity away from an R_{state} -fixed pair, multi-edge or all-iterate perturbation stability, or priority. The selector bound applies only to infinite certified strict branches, and the lower-ideal lemma stops after one step.

Reproducibility here means literal symbolic derivations and the exact arithmetic in (7.2)–(7.5). No code, data, numerical scan, computer algebra certificate, fitted parameter, or hardware experiment is needed or used. Every displayed inequality is a finite certificate that can be checked from the printed support vectors. A routine appendix, if supplied, may repeat arithmetic already explained but may not introduce a new hypothesis or computational claim.

The central message is therefore modest but exact. Positive collected supports plus strict certificates make a high-dimensional selector process translate along one diagonal ray. The finite wall count controls its transient selector word, the stationary phase has an affine recurrence, and reversal is an observable-span statement rather than an automatic global symmetry. The local margin lemma and the fixtures show both how the mechanism can be used and where it stops. Extending it to larger support actions or finite-fan compactifications would require new certificates, not a change of notation.

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